

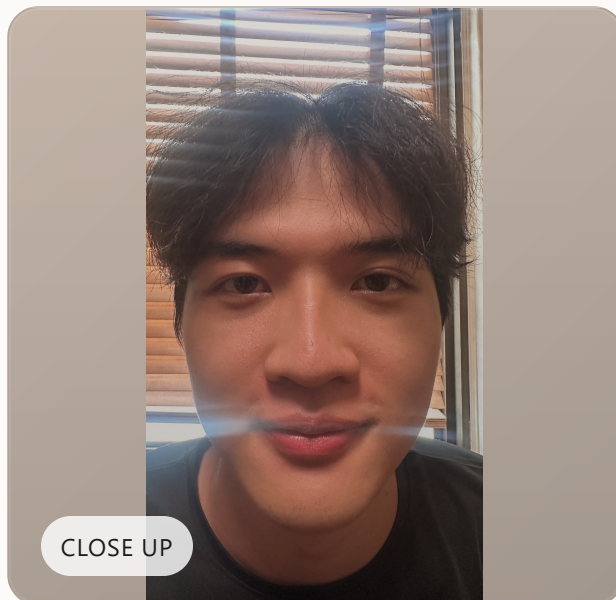
CS 180 • PROJECT 0

Perspective, Focal Length, and the Center of Projection

In this class, we want you to become friends with every pixel. The first step toward this goal is to become friends with your camera—e.g., the one in your smartphone. In particular, the goal of this project is to get you some intuitive understanding of the somewhat subtle relationship between perspective, focal length/zoom, and the center of projection.

Part 1: (Selfie) The Wrong Way vs. The Right Way

Take a picture of your friend (or yourself) from close up. You get a typical distorted selfie image. Now step back several feet from your subject, zoom in, and take a second picture. Try to get the face in the second photo to be the same size as in the first photo. If you've done things right, the second portrait should look much better than the first one.

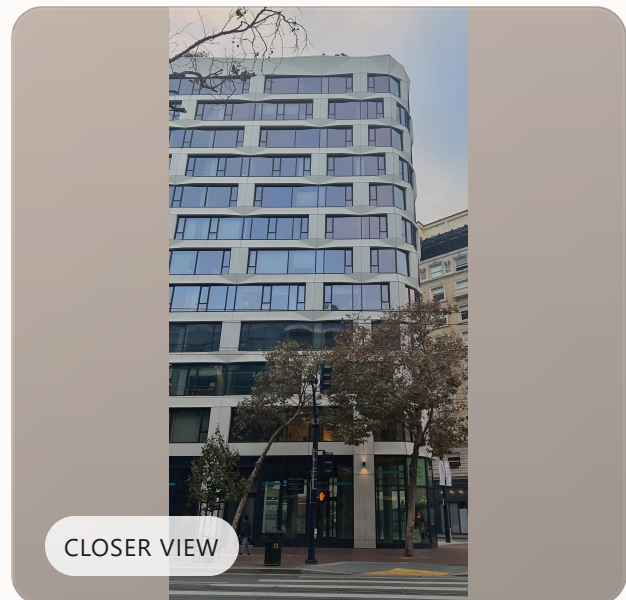
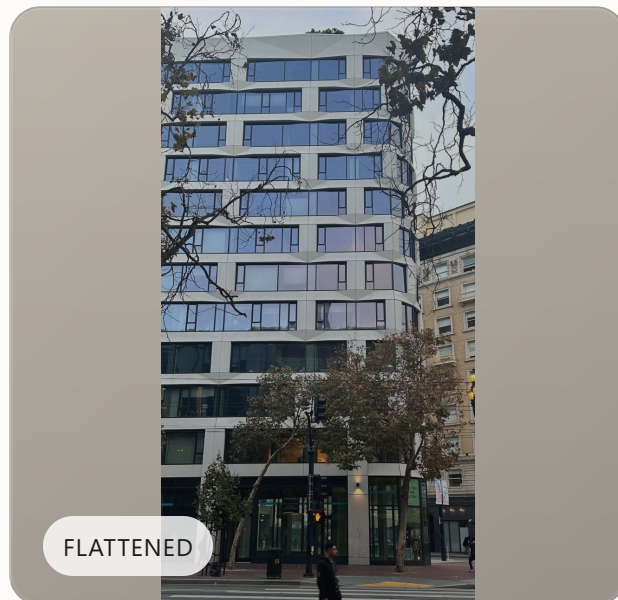


I moved farther away and zoomed in, so his face stayed about the same size but the background changed a lot. This made the picture look more normal because I was

not so close that the face got stretched and weird.

Part 2: Architectural Perspective Compression

Let's repeat the same procedure in reverse, for an urban scene. Pick an empty field with a nice building facade in the background (not totally frontal). Zoom in, and take a photo of the building. Now, walk toward the building, and take a second photo, now without zoom, such that the building in the two photos appears approximately the same size. The first picture should look flattened, or compressed, compared to the second.

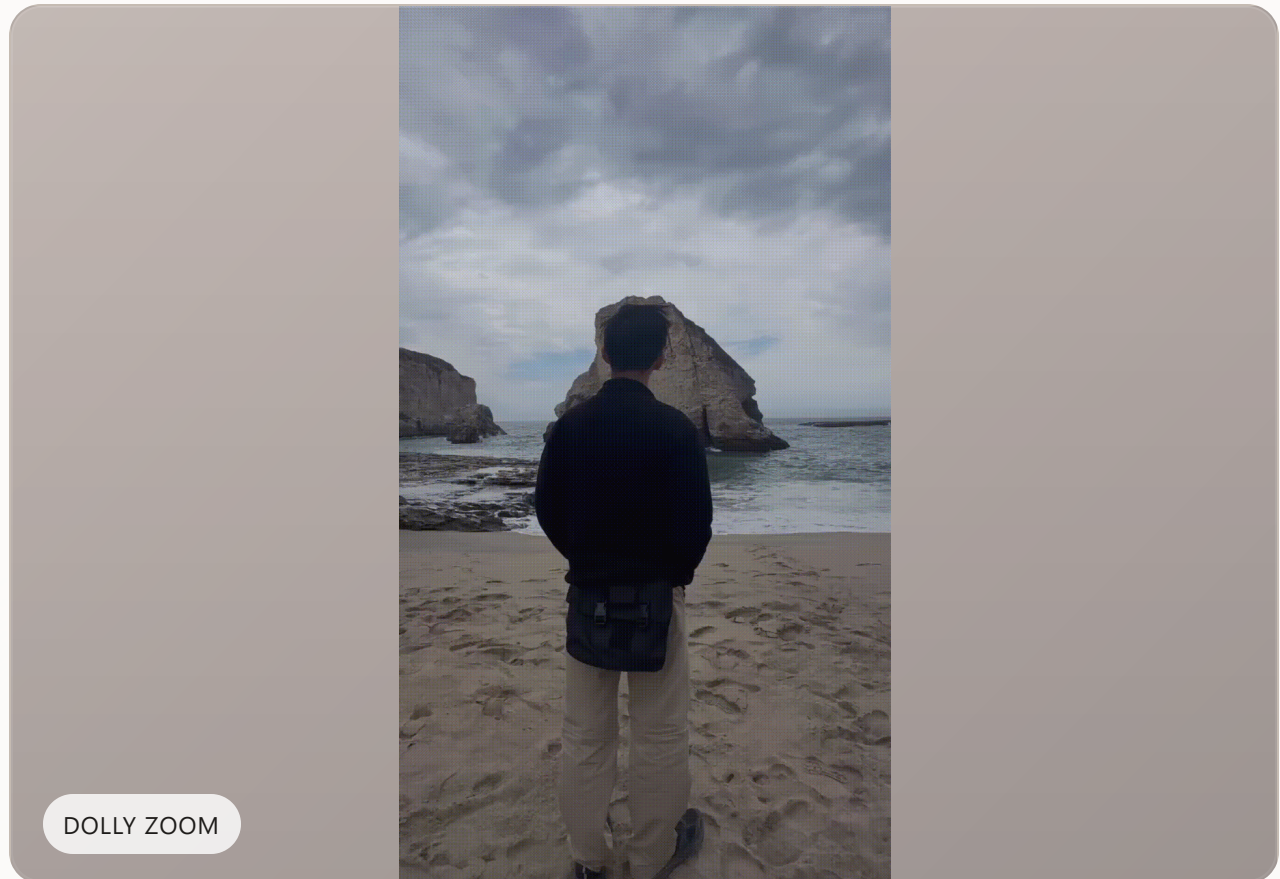


I zoomed in from farther away, which is basically like taking the photo from close up and then cropping it. Because I was still farther back, the building looked flatter and less deep. When I walked closer without zooming, the scene looked more stretched out and had more depth in front of it.

Part 3: The Dolly Zoom

This classic effect is a simultaneous move backward while zooming in. Choose a scene, take a series of stills as you move back and zoom in, and keep the central subject roughly

the same size. Combine the stills into an animated GIF to create a dolly zoom.



I moved backward while zooming in, so the main object stayed about the same size. At the same time, the background changed a lot, which made the scene feel like it was getting bigger or more dramatic while the subject stayed in place.